

# Manual of the Field-Free VC-MTJ Compact Model

A Verilog-A model capturing hybrid dynamics supports monte-carlo simulation in Cadence Virtuoso

**Version:** 1.0

**Release Date:** 2026/07/30

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| Version | Release Date | Description of Changes                                                                                                                                                                                                                                                                                                                                    | Contributor                                  |
|---------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| 1.0     | 2026/07/30   | Initial release.<br>1. Developed a probabilistic compact model for the field-free VC-MTJ using Verilog-A.<br>2. Implemented the evaluation of voltage-dependent resistance and stochastic switching.<br>3. Provided typical and Monte Carlo simulation configurations in Virtuoso.<br>4. Included comprehensive netlist and schematic testbench examples. | Yantong, Yaoru, Xuezhao, Hao Cai, and Qiming |
|         |              |                                                                                                                                                                                                                                                                                                                                                           |                                              |
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# **I. General Introduction**

Voltage-controlled magnetic anisotropy magnetic tunnel junctions (VC-MTJs) provide a voltage-driven approach to controlling the magnetic state of an MTJ. The field-free VC-MTJ considered in this model is intended for circuit-level evaluation of voltage-dependent resistance and stochastic state switching without requiring an externally applied magnetic field in the user testbench.

The VC\_MTJ package provides an experimentally calibrated compact model implemented in Verilog-A. The model describes the electrical response of the MTJ in its parallel (P) and antiparallel (AP) states and evaluates the probability of switching under an applied write pulse. The switching response depends on the voltage actually developed across the MTJ terminals and on the effective pulse duration seen by the device.

The model is designed for Cadence Spectre and Virtuoso simulations. It supports a typical model corner using nominal device parameters and a Monte Carlo corner for mismatch analysis. Both netlist-based and schematic-based example testbenches are included to demonstrate model integration and basic operation.

This is a probabilistic compact model rather than a full magnetization-dynamics solver. It does not numerically integrate the complete stochastic Landau-Lifshitz-Gilbert equation and does not provide a continuous magnetization trajectory. Instead, it provides a compact P/AP state description suitable for transient and Monte Carlo circuit simulation.

## II. Files Provided

### II.A Package Organization

```
0_VC_MTJ_Model/
├── README.md
├── model_file/
│   ├── vc_mtj_core.lib
│   ├── vc_mtj_core.mdl
│   └── vc_mtj_core.va
└── testbench/
    ├── schematic_based/
    │   ├── README.md
    │   └── tb/
    │       ├── schematic/
    │       └── adexl/
    │           └── vc_mtj_3t/
    └── netlist_based/
        ├── README.md
        ├── tb_typical.scs
        ├── run_tb_typical.sh
        ├── tb_mc.scs
        └── run_tb_mc.sh
```

### II.B Model Files

| File                              | Content                                                                                                                      |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <i>model_file/vc_mtj_core.lib</i> | Defines model parameters and the mtj_t/ mtj_mc model corners, then includes the Spectre wrapper.                             |
| <i>model_file/vc_mtj_core.mdl</i> | Defines the external three-terminal Spectre subcircuit vc_mtj_3t and maps model-library parameters to the Verilog-A module.  |
| <i>model_file/vc_mtj_core.va</i>  | Implements voltage-dependent resistance, probabilistic switching, internal random-number generation, and P/AP state updates. |

Users normally include the .lib file and instantiate vc\_mtj\_3t, direct modification of the .mdl or .va files is not required for standard use.

### II.C Testbench Files

| File or Directory                             | Description                                           |
|-----------------------------------------------|-------------------------------------------------------|
| <i>testbench/netlist_based/tb_typical.scs</i> | Typical-corner transient example using section=mtj_t. |

| File or Directory                                | Description                                                                                |
|--------------------------------------------------|--------------------------------------------------------------------------------------------|
| <i>testbench/netlist_based/run_tb_typical.sh</i> | Runs the typical netlist testbench and writes the typical simulation results.              |
| <i>testbench/netlist_based/tb_mc.scs</i>         | Mismatch Monte Carlo example using section=mtj_mc.                                         |
| <i>testbench/netlist_based/run_tb_mc.sh</i>      | Runs the Monte Carlo netlist testbench and writes the MC simulation results.               |
| <i>testbench/schematic_based/vc_mtj_3t/</i>      | Virtuoso device cell containing the three-terminal symbol and model interface.             |
| <i>testbench/schematic_based/tb/schematic/</i>   | Virtuoso schematic testbench for the VC-MTJ compact model.                                 |
| <i>testbench/schematic_based/tb/adexl/</i>       | Saved ADE XL setup supporting both typical transient and mismatch Monte Carlo simulations. |
| README.md files                                  | Provide quick-start instructions for the package root and each testbench style.            |

Running the supplied netlist testbenches produces result directories and Spectre log files in *testbench/netlist\_based/*. These outputs are generated locally and are not required model-source files.

| Testbench   | Generated Output                                   |
|-------------|----------------------------------------------------|
| Typical     | <i>psf_typical/</i> and <i>spectre_typical.log</i> |
| Monte Carlo | <i>psf_mc/</i> and <i>spectre_mc.log</i>           |

### III. Model Interface

#### III.A Device Symbol and Pins

The external Spectre subcircuit is named `vc_mtj_3t`. Its port order is `p`, `n`, `state`. The corresponding Verilog-A module is `VC_MTJ_core(p, n, state)`.

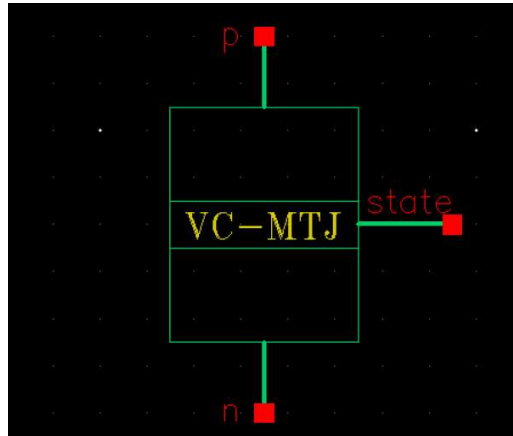


Fig. 1. Virtuoso symbol of the `vc_mtj_3t` compact model.

| Pin   | Type                       | Description                                                                     |
|-------|----------------------------|---------------------------------------------------------------------------------|
| p     | Analog electrical terminal | Positive reference terminal of the MTJ. The model defines $V_{MTJ} = V(p,n)$ .  |
| n     | Analog electrical terminal | Negative reference terminal of the MTJ.                                         |
| state | Observation output         | Outputs the MTJ resistance state as a smoothed voltage: 0V for P and 1V for AP. |

The state pin is provided for observing the internal MTJ state. It should not be driven by an external voltage source or connected as an electrical control input.

#### III.B Initial State

The initial resistance state is selected using the instance parameter `Init`.

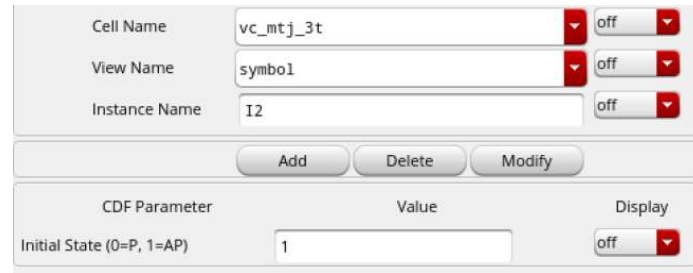
| Setting | Type              | Description |
|---------|-------------------|-------------|
| Init=0  | Parallel (P)      | 0 V         |
| Init=1  | Antiparallel (AP) | 1 V         |

For a Spectre netlist, the initial state is assigned in the instance statement:

```
I0 (VMTJ net1 State) vc_mtj_3t Init=1
```

In a Virtuoso schematic, the same parameter is set through the instance properties or the component

CDF, depending on the final packaged cell configuration.



| CDF Parameter             | Value | Display |
|---------------------------|-------|---------|
| Initial State (0=P, 1=AP) | 1     | off     |

Fig. 2. Setting the Init parameter in Virtuoso.

### III.C Electrical and Switching Behavior

The compact model continuously contributes current between p and n according to the voltage across the MTJ and its current P/AP state. The P-state resistance, AP-state resistance, and tunnel magnetoresistance are bias dependent; therefore, the model should not be treated as a fixed two-resistor element.

During a write operation, the model evaluates the pulse that is actually present across the MTJ terminals. The switching probability is determined from the effective pulse voltage, pulse duration, current state, and calibrated device parameters. A stochastic decision then either toggles the state or leaves it unchanged. The state output and electrical resistance update consistently after the write event.

The user interface remains compact: circuit designers provide terminal excitation and the initial state, while the detailed resistance and switching calculations are handled internally by the model.

## IV. Model Usage

### IV.A Netlist-Based Typical Simulation

The typical example is provided in *0\_VC\_MTJ\_Model/testbench/netlist\_based/*. It uses the *mtj\_t* model corner and performs a standard transient simulation.

➤ **Model inclusion and instantiation**

```
include "../model_file/vc_mtj_core.lib" section=mtj_t
I0 (VMTJ net1 State) vc_mtj_3t Init=1
```

In the provided example, the MTJ is initialized in the AP state and driven by a 2.4V pulse over a 10ns transient simulation. The complete stimulus and analysis settings are defined in *0\_VC\_MTJ\_Model/testbench/netlist\_based/tb\_typical.scs*.

➤ **Run procedure**

1. Open a shell in the release package or change to the netlist-based testbench directory.
2. Run the supplied script.
3. Check *spectre\_typical.log* for completion and open the generated PSF results in ViVA.

```
cd 0_VC_MTJ_Model/testbench/netlist_based
./run_tb_typical.sh
```

The script writes the results to *psf\_typical/* and the simulator output to *spectre\_typical.log*. The example is intended to verify model compilation, transient execution, and basic state/resistance behavior.

### IV.B Netlist-Based Monte Carlo Simulation

The Monte Carlo example is provided in the same directory. It uses the *mtj\_mc* model corner and performs a mismatch Monte Carlo simulation.

➤ **Model inclusion and instantiation**

```
include "../model_file/vc_mtj_core.lib" section=mtj_mc
I0 (VMTJ net1 State) vc_mtj_3t Init=1
```

➤ **Monte Carlo analysis**

```
mc1 montecarlo
numruns=1
seed=24680
variations=mismatch
donominal=no
{
    tran stop=10 ns
}
```

Both the `mtj_mc` model corner and `variations=mismatch` are required to enable mismatch sampling. The complete settings are defined in `tb_mc.scs`.

➤ **Run procedure**

1. Open a shell in the release package or change to the netlist-based testbench directory.
2. Run the supplied script.
3. Check `spectre_mc.log` for completion and open the generated PSF results in ViVA.

```
cd 0_VC_MTJ_Model/testbench/netlist_based
```

```
./run_tb_mc.sh
```

The results are written to `psf_mc/` and the simulator output to `spectre_mc.log`.

## IV.C Schematic-Based Typical Simulation

The schematic-based package is located in `0_VC_MTJ_Model/testbench/schematic_based/`. The saved `tb/adexl` view provides a unified ADE XL setup for both typical transient and mismatch MC simulations. The same `tb/schematic` testbench and saved output definitions are used in both cases.

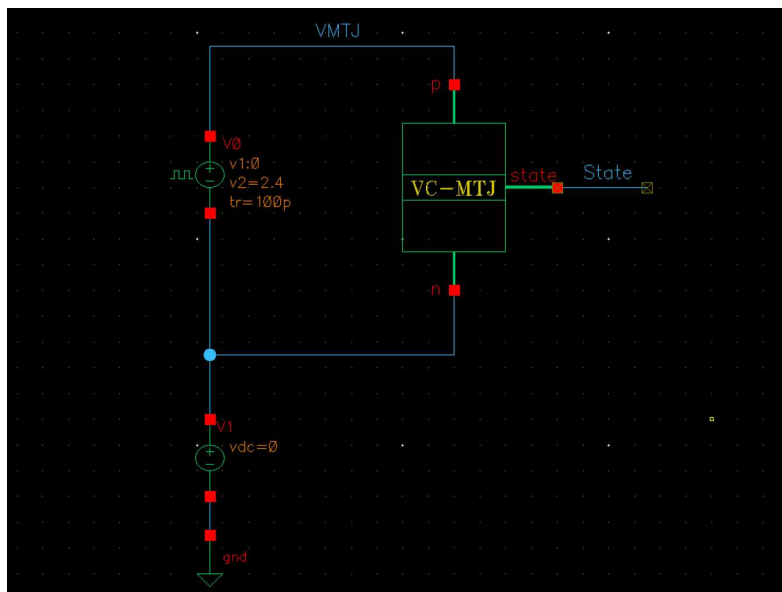


Fig. 3. Virtuoso schematic-based testbench for the VC-MTJ compact model.

➤ **Opening the saved setup**

1. Add `testbench/schematic_based/` as a Virtuoso library and open the saved `tb/adexl` view.
2. Update the library path to the local location of `model_file/vc_mtj_core.lib` when necessary.
3. Confirm that the configured transient analysis and the saved outputs are enabled.

➤ **Typical transient run**

1. Select **Single Run, Sweeps and Corners** as the run mode.
2. In **Corners Setup**, enable the typical corner, which uses the `mtj_t` model section.
3. Run the configured transient analysis and inspect the saved waveforms in ViVA.

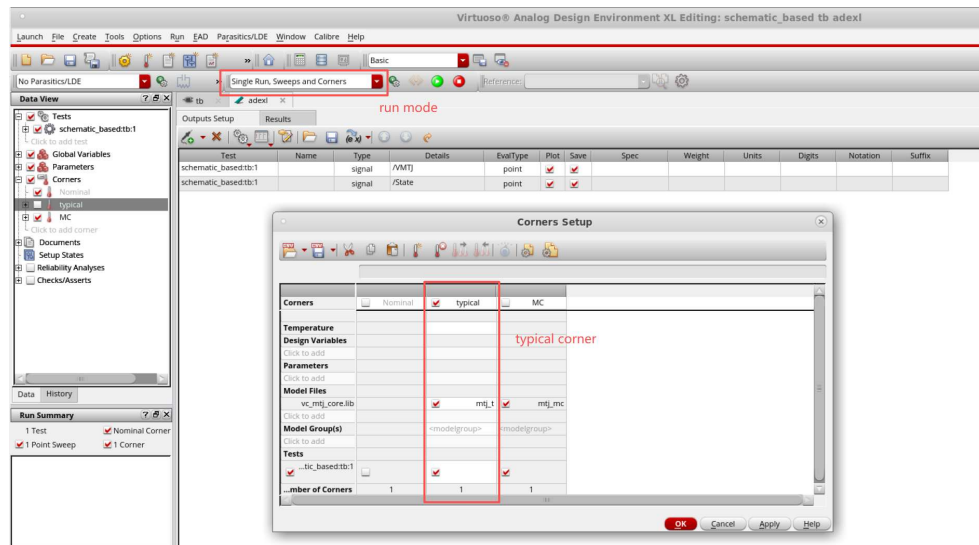


Fig. 4. ADE XL setup for the schematic-based typical transient simulation.

## IV.D Schematic-Based Monte Carlo Simulation

The same saved *tb/adexl* view is used for mismatch Monte Carlo simulation. The schematic, transient analysis, and output definitions remain unchanged; only the run mode, active corner, and Monte Carlo settings are changed.

### ➤ Monte Carlo run

1. Select Monte Carlo Sampling as the run mode.
2. In Corners Setup, enable the MC corner, which uses the *mtj\_mc* model section.
3. In the Monte Carlo settings, select Mismatch as the statistical variation. Set the required number of points and Monte Carlo seed before running the simulation.
4. Run the Monte Carlo simulation and inspect the saved results.

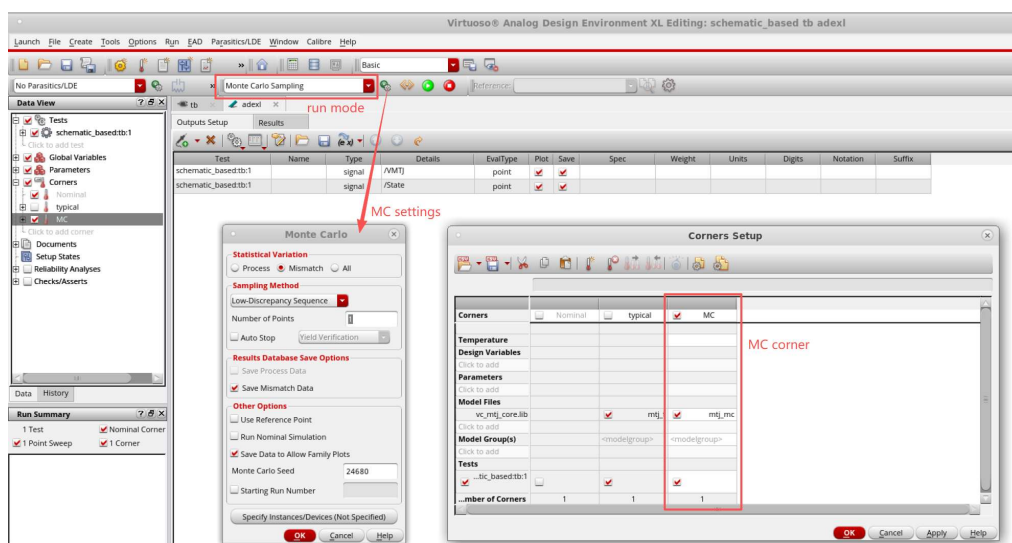


Fig. 5. ADE XL setup for the schematic-based mismatch Monte Carlo simulation.

## V. Reference Simulation Results

This section provides reference outputs for confirming that the released model and testbenches are installed and running correctly. The switching sequence may vary with the random sequence and Monte Carlo realization. Therefore, the figures should be interpreted as reference examples rather than a required pulse-by-pulse state patterns.

### V.A Typical Switching Example

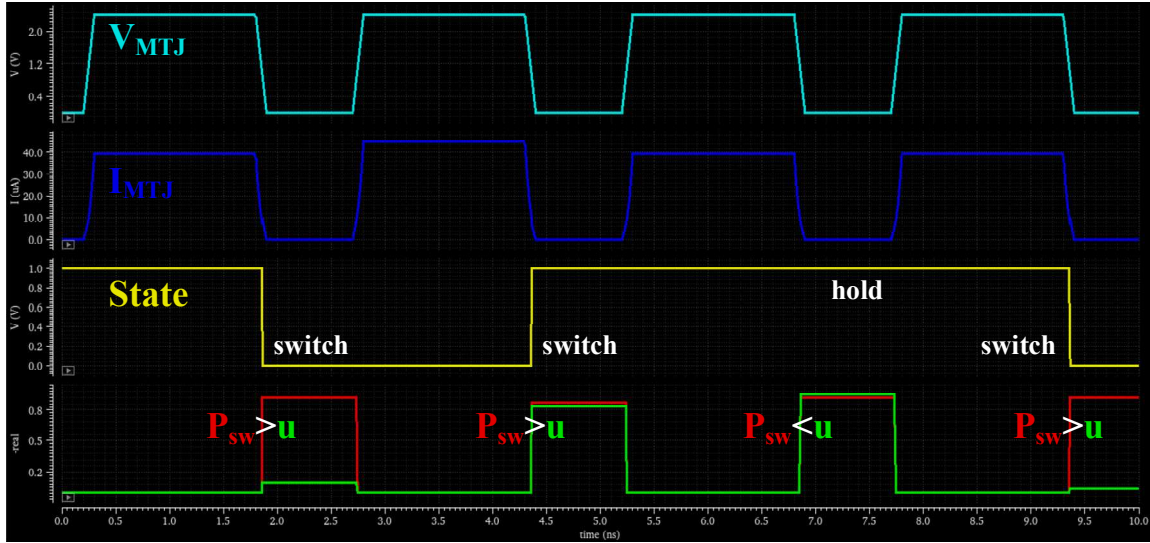


Fig. 6. Typical-corner transient response of the VC-MTJ compact model.

Fig. 6 shows the applied MTJ voltage  $V_{MTJ}$ , MTJ current  $I_{MTJ}$ , state output, switching probability  $P_{sw}$ , and pseudo-random number  $u$ . For each write pulse, the MTJ state switches when  $u < P_{sw}$  and remains unchanged otherwise. The MTJ current varies accordingly with the P/AP state.

### V.B Monte Carlo Example

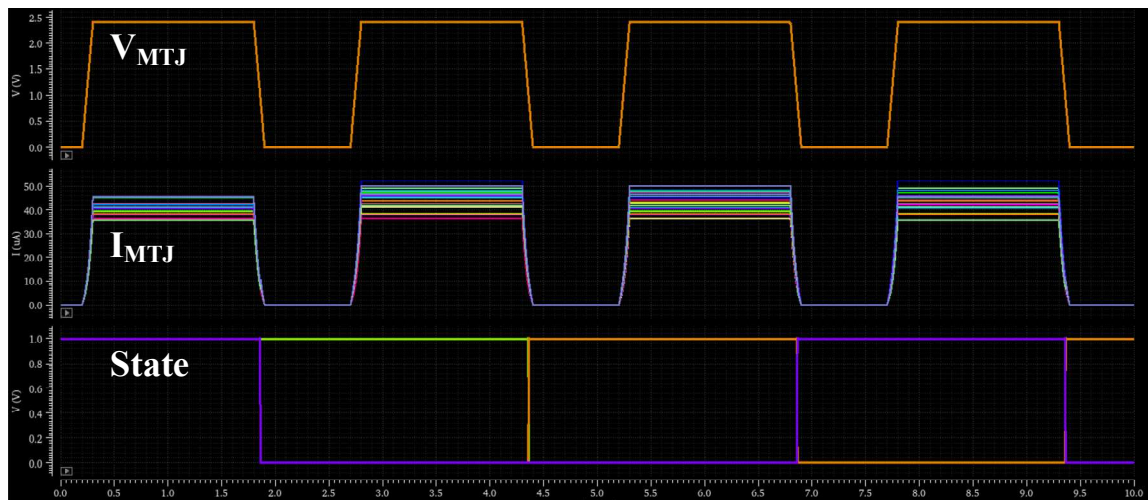


Fig. 7. Monte Carlo transient responses of the VC-MTJ compact model.

Each Monte Carlo run represents a different device realization. Under identical input pulses, variations are observed in the MTJ current and probabilistic switching sequence, as shown in Fig. 7.

### V.C Voltage-Dependent Resistance

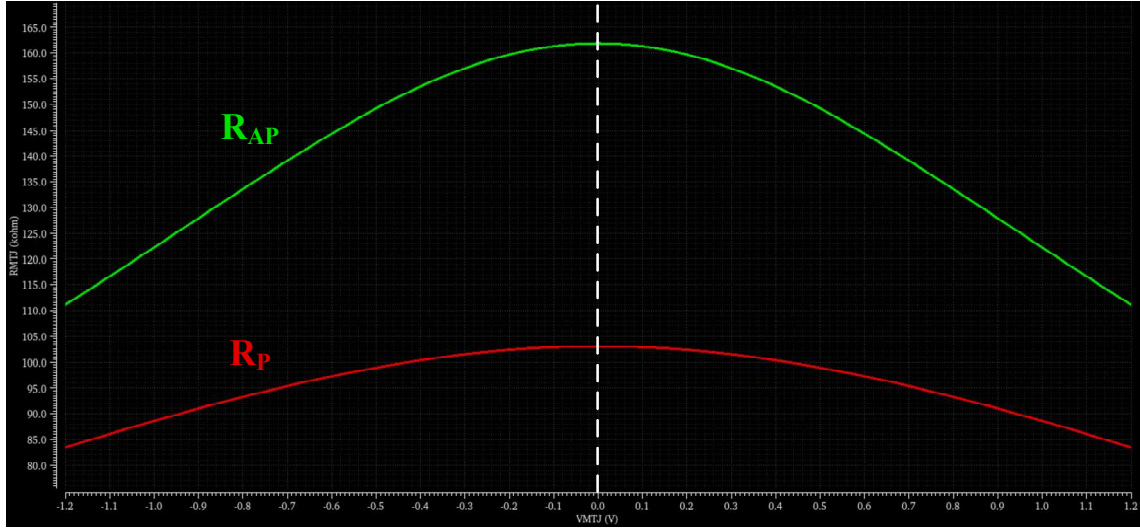


Fig. 8. Voltage-dependent P- and AP-state resistances of the VC-MTJ compact model.

The DC sweep in Fig. 8 illustrates the voltage-dependent resistance behavior implemented in the compact model. Both  $R_P$  and  $R_{AP}$  decrease as the magnitude of the applied MTJ voltage increases.

## VI. Model Parameters

The main user-accessible model settings are summarized below. Other calibrated parameters are defined in *model\_file/vc\_mtj\_core.lib* and are not intended for routine modification.

| Setting | Options        | Description                                                                   |
|---------|----------------|-------------------------------------------------------------------------------|
| Init    | 0, 1           | Sets the initial MTJ state: 0 for P and 1 for AP.                             |
| mtj_t   | Typical corner | Uses the nominal model parameters.                                            |
| mtj_mc  | MC corner      | Enables the statistical model parameters for mismatch Monte Carlo simulation. |

## VII. Usage Notes and Model Limitations

- The model is an experimentally calibrated probabilistic compact model for circuit-level simulation and does not solve the full stochastic LLG equation.
- The switching response is calibrated over a fitted voltage range of 1.6–2.4V. Results outside this range should not be interpreted as independently validated physical predictions.
- The calibrated switching model requires an effective pulse-top duration of at least 0.7ns.
- The switching response is evaluated using the voltage developed across the MTJ terminals, which may differ from the programmed source voltage in circuits containing access devices, interconnect resistance, or other loading.
- The calibrated resistance and switching parameters should not be modified independently without model recalibration. The supplied testbenches are functional examples, and application-specific pulse conditions and read margins should be verified separately.